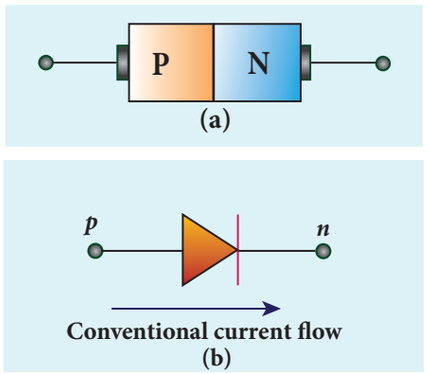


Figure 10.11 *p-n* junction diode
(a) Schematic representation
(b) Circuit symbol

Biasing a diode

Biasing means providing external energy to charge carriers to overcome the barrier potential and make them move in a particular direction. The charge carriers can either move towards the junction or away from the junction. **The external voltage applied to the *p-n* junction is called bias voltage**. Depending on the polarity of the external source to the *p-n* junction, we have two types of biasing:

i) Forward bias
ii) Reverse bias

i) Forward bias

If the positive terminal of the external voltage source is connected to the *p*-side and the negative terminal to the *n*-side, it is called forward bias as shown in Figure 10.12. The application of a forward bias potential pushes electrons in the *n*-side and the holes in the *p*-side towards the junction. This initiates the recombination with the ions near the junction which in turn reduces the width of the depletion region and hence the barrier potential.

The electron from the *n*-side is now accelerated towards the *p*-side as it experiences a reduced barrier potential at the junction. In addition, the accelerated electrons experience a strong attraction by the positive potential applied to the *p*-side. This results in the movement of electrons in the *n*-side towards the *p*-side and similarly, holes in the *p*-side towards the *n*-side. When the applied voltage is increased, the width of the depletion region and hence the barrier potential are further reduced. This results in a large number of electrons passing through the junction resulting in an exponential rise in current through the junction.

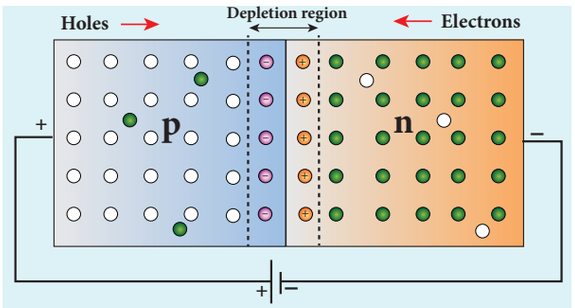


Figure 10.12 Schematic representation of a *p-n* junction diode under forward bias

ii) Reverse bias

If the positive terminal of the battery is connected to the *n*-side and the negative terminal to the *p*-side, the junction is said to be reverse biased as shown in Figure 10.13.

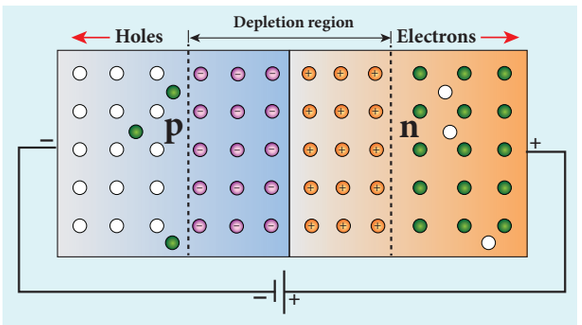


Figure 10.13 Schematic representation of a *p-n* junction diode under reverse bias

As the positive terminal is connected to the *n*-type material, the electrons in the *n*-side are attracted towards the positive

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